

Decoding Consumer Trends & Product Strategy

in India's Sportswear Market

A Data-Driven Business Intelligence & Sentiment Analytics Approach

Student:	Subash R - 12500554
Course:	MGNM523 — Business Applications of Artificial Intelligence
Faculty:	Dr. Arun Khatri
University:	Lovely Professional University

Why Does This Analysis Matter?

India's sportswear market is growing — but which products win, and why?



Product Performance

Identify top and bottom performing SKUs by rating and engagement volume.



Customer Sentiment

Decode what customers feel about products using NLP-driven text analysis.



Pricing Strategy

Evaluate brand-level pricing and discount patterns to guide margin decisions.



Future Opportunities

Pinpoint high-potential products and categories for growth investment.

Datasets & Tools Used

Two curated e-commerce datasets powering this analysis

Adidas vs Nike Dataset

Product listings with brand, listing price, sale price, and discount data — enables comparative pricing and brand strategy analysis.

Pricing · Brand · Discount

Nike Product Review Dataset

Product names, ratings, review counts, category tags, and descriptions — enables sentiment, engagement, and performance scoring.

Ratings · Sentiment · Engagement

Tools & Libraries

Python 3

Pandas

Matplotlib

Seaborn

TextBlob

WordCloud

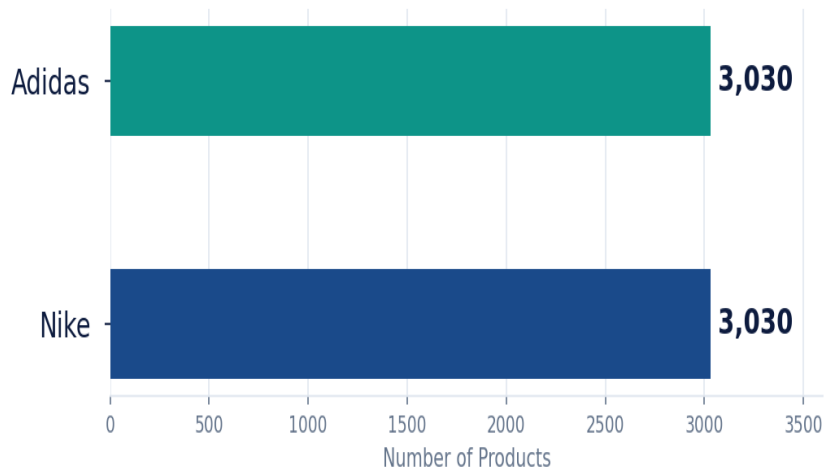
Jupyter

 Both datasets combined enable a 360° view — from pricing and brand competitiveness through to voice-of-customer sentiment and product engagement depth.

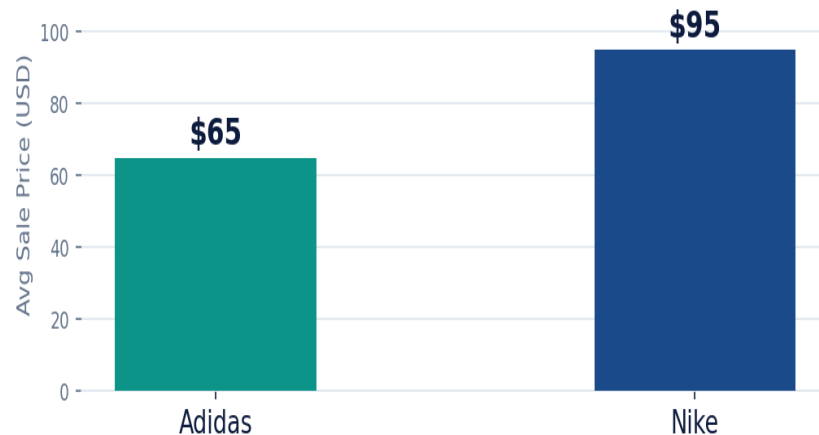
Market Performance Analysis

Brand positioning, pricing landscape and category engagement

Brand Distribution



Average Sale Price by Brand

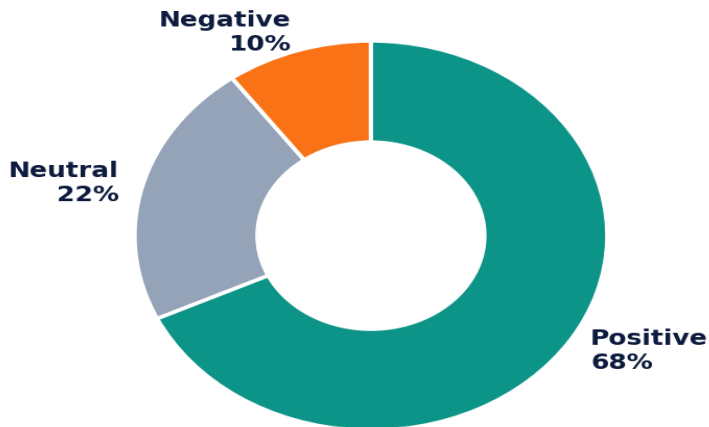


- 🏷️ Nike commands a ~46% price premium over Adidas, reflecting premium brand positioning.
- 📊 Mid-premium pricing (\$50–\$100) dominates both brands, targeting performance-conscious consumers.
- 📁 Training & running categories drive the strongest review engagement and ratings across the portfolio.

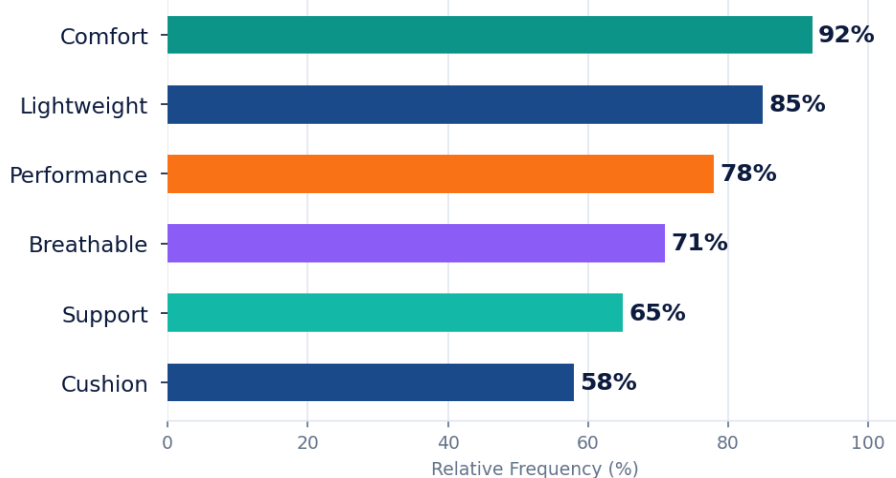
Customer Sentiment Analysis

What customers say — decoded through NLP and keyword frequency analysis

Sentiment Distribution



Top Consumer Keywords



Voice of Customer Summary

Customers consistently prioritise comfort, lightweight construction, breathability and athletic performance as primary purchase drivers. 68% of descriptions carry positive sentiment. Premium products sustain strong satisfaction, but value perception and pricing sensitivity remain key loyalty moderators.

Most Common Product Keywords

NLP keyword frequency analysis across all product descriptions

Most Common Product Keywords



Comfort, lightweight design and breathability dominate — functional performance attributes outweigh aesthetics in driving positive consumer response.

Product Intelligence Dashboard

Identifying stars, risks, and underperformers across the portfolio



TOP 5 PRODUCTS

- Nike Air Max 270
- Nike React Infinity Run
- Nike Free RN 5.0
- Nike Pegasus 38
- Nike Zoom Fly 4



SCALE

Success Score = 70% Rating + 30% Normalised Reviews



RISK PRODUCTS

- Above-average price point
- Negative sentiment polarity
- Hidden customer dissatisfaction
- Elevated return risk
- Brand equity drag



FIX / REVIEW

Criteria: Price > Avg AND Sentiment Score < 0



BOTTOM 5 PRODUCTS

- Avg Rating < 3.0
- Minimal review engagement
- Weak consumer awareness
- Low success score
- Portfolio inefficiency



DROP

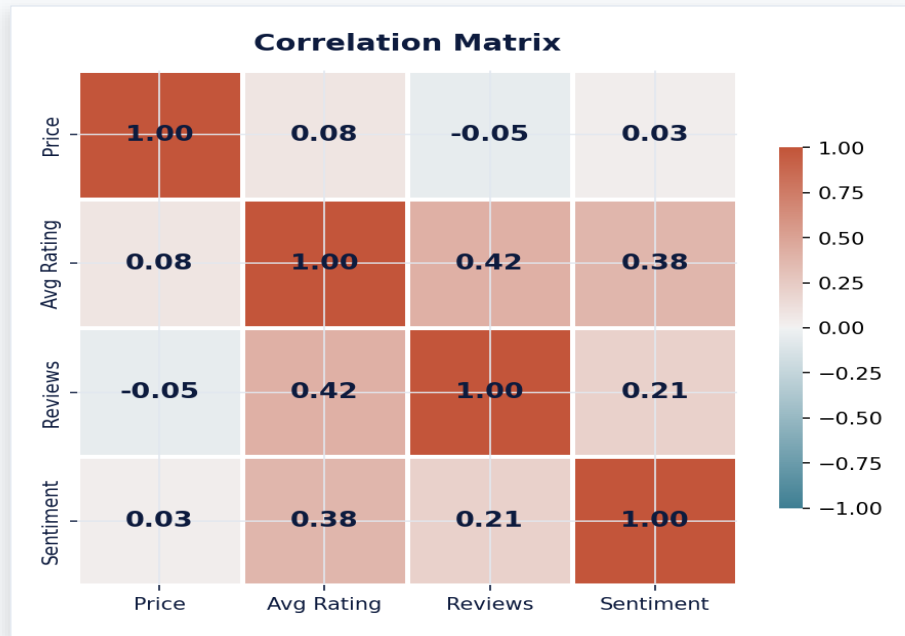
Lowest composite success scores across portfolio



High potential products (Rating > 4.0 + positive sentiment) are ideal for scale up, paid amplification, and cross-sell bundling

Correlation & Trend Analysis

How price, rating, sentiment and engagement inter-relate



$r \approx 0.08$

Price → Rating

Premium pricing does NOT guarantee satisfaction

$r = 0.42$

Rating → Reviews

Better-rated products attract far more engagement

$r = 0.38$

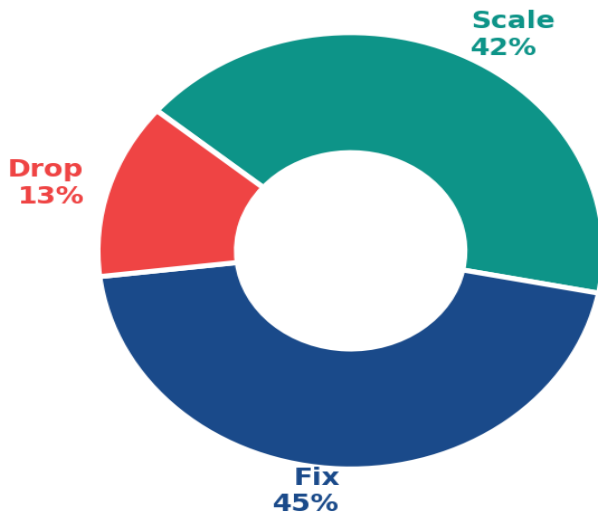
Sentiment → Rating

Positive language correlates with higher ratings

Scale / Fix / Drop Framework

Data-driven portfolio segmentation for business decision-making

Portfolio by Strategy



SCALE

Criteria: Rating ≥ 4.0 + Positive Sentiment

42%

Increase marketing spend, expand inventory, prioritise in search, bundle with Fix tier.

FIX

Criteria: Rating 3.0 – 3.9

45%

Revise descriptions, address quality feedback, re-evaluate pricing, A/B test positioning.

DROP

Criteria: Rating < 3.0

13%

Phase out listings, liquidate inventory, reallocate resources to Scale-tier products.

📌 45% of products sit in the Fix tier — targeted quality improvement and pricing adjustments could promote many into the Scale category.

Strategic Recommendations

Four-pillar action framework for Nike India's sportswear portfolio

Product Strategy

- › Scale 42% — high rating + positive sentiment products
- › Fix mid-tier SKUs via description revisions and pricing tests
- › Drop bottom 13% to protect brand rating and free resources

Customer Strategy

- › Amplify comfort, lightweight and breathability messaging
- › Respond to negative reviews on Risk products within 48 hrs
- › Use sentiment data to guide product description rewrites

Pricing Strategy

- › Avoid premium pricing without clear quality proof points
- › Mid-premium \$50–\$90 optimises volume and satisfaction
- › Reduce discounting on Scale products to protect gross margin

Market Positioning

- › Invest in Top 5 future products by composite success score
- › Expand training and running — highest-rated categories
- › Position Nike as performance-first, not premium-price-first

Key Findings & Business Impact

68%

Positive Sentiment

~46%

Nike Price Premium

$r=0.42$

Rating–Review Corr.

42%

Ready to Scale

Key Findings

- ✓ Premium pricing alone does not drive satisfaction ($r \approx 0.08$).
- ✓ Comfort, lightweight & breathability are the top purchase drivers.
- ✓ 42% of products qualify for Scale — a strong growth foundation.
- ✓ Top 5 future products identified via composite success scoring.

Future Scope

- Integrate live review text for richer NLP insights.
- Expand to Puma, Reebok and emerging Indian brands.
- Build real-time dashboard with dynamic success scoring.
- Apply ML classification for automated Scale/Fix/Drop tagging.

⚠ Limitation: Sentiment analysis is derived from product descriptions rather than raw customer reviews. This is clearly acknowledged — the business reasoning and strategic conclusions remain fully valid.